

VortexTech Cybersecurity Internship

Week 2 Report

Hands-on with Basic Security Tools

Name: Aiman Arjmand

Track: Cybersecurity Internship

Week: 2

1. Objective

The objective of this task was to gain practical experience with two fundamental cybersecurity concepts: password security and network reconnaissance. A Python-based password strength checker was used to evaluate password complexity, while Nmap was used to scan localhost for open ports. This exercise helped demonstrate how weak passwords and exposed network services can increase security risks.

2. Password Strength Checker

Overview

A Python program was used to evaluate password strength based on multiple security criteria. The program analyzes user-entered passwords and provides a strength rating along with suggestions for improvement.

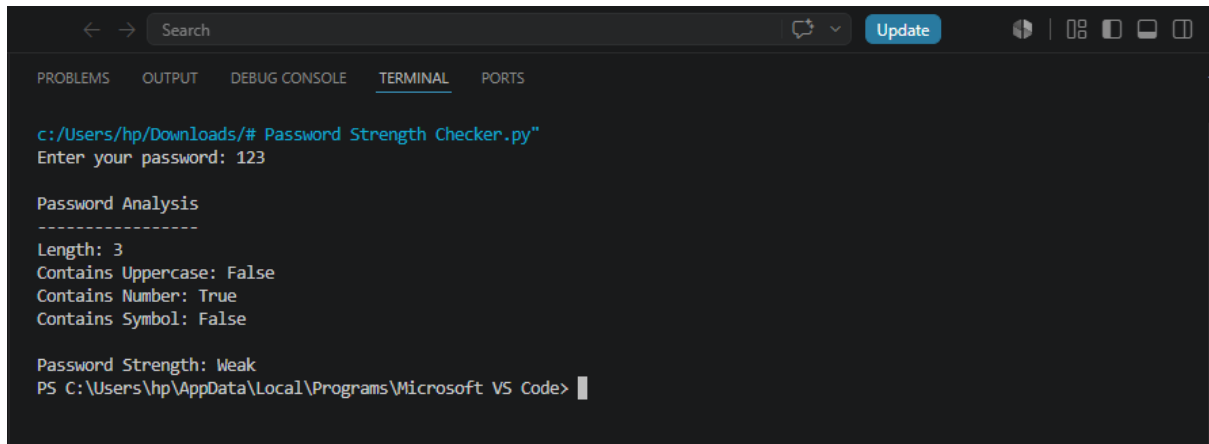
Password Evaluation Criteria

- Minimum length of 8 characters
- Contains uppercase letters
- Contains lowercase letters
- Contains numbers
- Contains special characters
- Checks against common passwords
- Provides feedback for improving weak passwords

Example Outputs

Password	Strength	Reason
password	Very Weak	Common password
123456	Very Weak	Common password
Cyber2026	Medium	Missing special character
Cyber@2026	Strong	Meets all requirements

Weak Password:



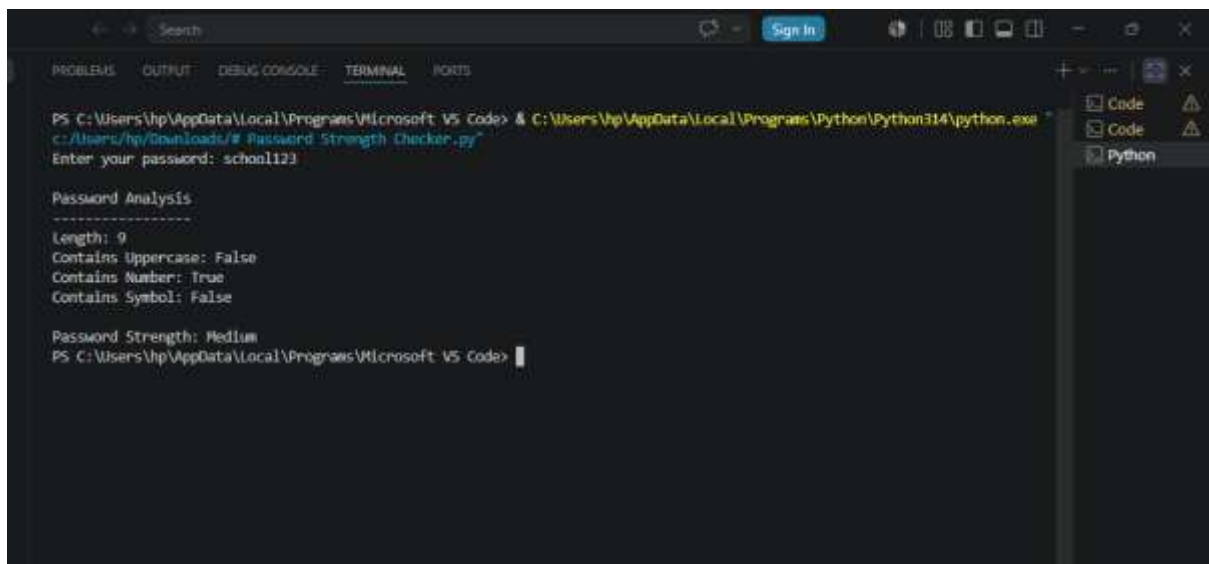
A screenshot of the Visual Studio Code terminal window. The terminal shows the execution of a Python script 'Password Strength Checker.py' located at 'c:/Users/hp/Downloads/'. The user is prompted to 'Enter your password:' and enters '123'. The script then performs a 'Password Analysis' and outputs the following details: Length: 3, Contains Uppercase: False, Contains Number: True, and Contains Symbol: False. The final result is 'Password Strength: Weak'. The terminal prompt is 'PS C:\Users\hp\AppData\Local\Programs\Microsoft VS Code>'.

```
c:/Users/hp/Downloads/# Password Strength Checker.py"
Enter your password: 123

Password Analysis
-----
Length: 3
Contains Uppercase: False
Contains Number: True
Contains Symbol: False

Password Strength: Weak
PS C:\Users\hp\AppData\Local\Programs\Microsoft VS Code>
```

Medium Password:



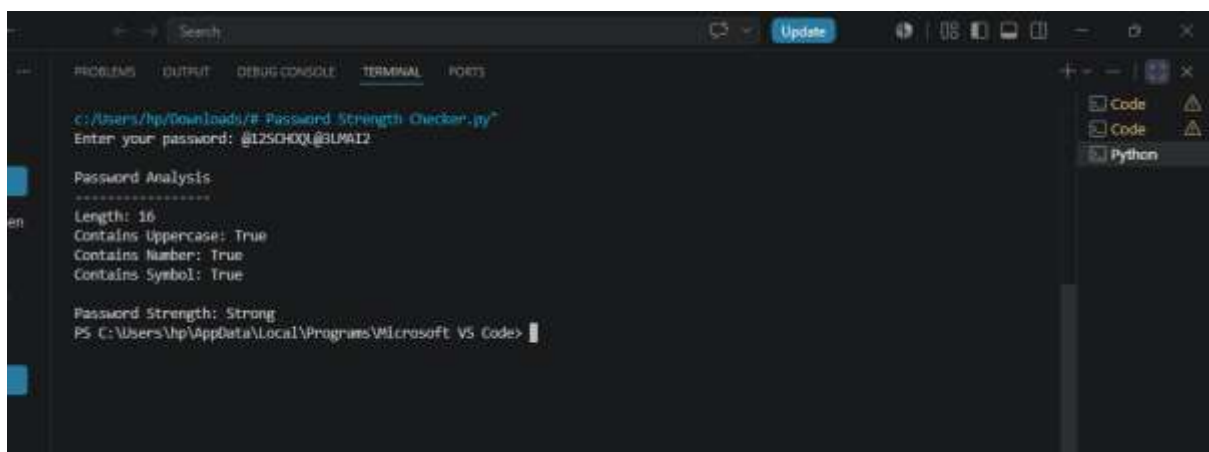
A screenshot of the Visual Studio Code terminal window. The terminal shows the execution of a Python script 'Password Strength Checker.py' located at 'c:/Users/hp/Downloads/'. The user is prompted to 'Enter your password:' and enters 'school123'. The script then performs a 'Password Analysis' and outputs the following details: Length: 9, Contains Uppercase: False, Contains Number: True, and Contains Symbol: False. The final result is 'Password Strength: Medium'. The terminal prompt is 'PS C:\Users\hp\AppData\Local\Programs\Microsoft VS Code>'.

```
PS C:\Users\hp\AppData\Local\Programs\Microsoft VS Code> & C:\Users\hp\AppData\Local\Programs\Python\Python314\python.exe "
c:/Users/hp/Downloads/# Password Strength Checker.py"
Enter your password: school123

Password Analysis
-----
Length: 9
Contains Uppercase: False
Contains Number: True
Contains Symbol: False

Password Strength: Medium
PS C:\Users\hp\AppData\Local\Programs\Microsoft VS Code>
```

Strong Password:



A screenshot of the Visual Studio Code terminal window. The terminal shows the execution of a Python script 'Password Strength Checker.py' located at 'c:/Users/hp/Downloads/'. The user is prompted to 'Enter your password:' and enters '@12SCHOOL@3UMA12'. The script then performs a 'Password Analysis' and outputs the following details: Length: 16, Contains Uppercase: True, Contains Number: True, and Contains Symbol: True. The final result is 'Password Strength: Strong'. The terminal prompt is 'PS C:\Users\hp\AppData\Local\Programs\Microsoft VS Code>'.

```
c:/Users/hp/Downloads/# Password Strength Checker.py"
Enter your password: @12SCHOOL@3UMA12

Password Analysis
-----
Length: 16
Contains Uppercase: True
Contains Number: True
Contains Symbol: True

Password Strength: Strong
PS C:\Users\hp\AppData\Local\Programs\Microsoft VS Code>
```

3. Port Scanning Using Nmap

Tool Used

Nmap (Network Mapper) was used to scan localhost (127.0.0.1) for open TCP ports.

Command Executed

nmap localhost

Output

A screenshot of a Linux desktop environment with a blue background. A terminal window is open in the center, displaying the output of the 'nmap localhost' command. The terminal text shows the Nmap version (7.99), the scan target (localhost), and the results indicating that all 1000 scanned ports are closed. The desktop has icons for 'HOME', 'File System', 'Trash', and a file named 'sf_CTF-sha...'. The system clock in the top right corner shows 12:41.

Scan Results

Port	Status	Service
None	Closed	No services detected

Analysis

The Nmap scan of localhost showed that all 1000 commonly scanned TCP ports were closed. This indicates that no services were actively listening for incoming network connections. Since no open ports were detected, the

system presents a smaller attack surface, reducing the likelihood of network-based attacks.

4. Security Reflection

Weak passwords are one of the most common causes of unauthorized account access because they can be easily guessed or cracked using dictionary and brute-force attacks. Using long, unique passwords with a combination of uppercase letters, lowercase letters, numbers, and special characters greatly improves security.

Open ports allow network communication with running services. While some services require open ports to function, unnecessary or unsecured open ports can provide attackers with potential entry points into a system. Regularly reviewing open ports and disabling unused services helps reduce security risks.

This task demonstrated that both password security and proper network configuration are essential components of a secure system. Strong authentication combined with a minimal attack surface significantly improves overall cybersecurity.

5. Conclusion

This internship task provided practical experience with password strength evaluation and basic network scanning. It highlighted the importance of creating strong passwords and minimizing exposed network services. These are fundamental cybersecurity practices that help protect systems against unauthorized access and common attacks.